A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 37

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** The time taken for half of the atoms of a radioactive isotope to decay (activity halves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	1

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 32 / 8 = 4 half-lives **1** 6400 / 2^4 = 400 Bq **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Bq) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** Short half-life isotopes (I-131) release all their radiation quickly: dangerous but only briefly so short-term precautions (potassium iodide tablets, evacuation) are sufficient **1** Very long half-life isotopes (U-238) emit only small amounts of radiation in any period so the level of danger is low **1** Decades-long half-lives combine a high rate of radiation release with persistence for most or all of a human lifetime, so short-term protection is inadequate (land contaminated for generations) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	3
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** Alpha particles have very low penetrating power and are absorbed by clothing / dead skin so external sources cause little exposure **1** Inside the body the alpha particles are absorbed in a very small mass of tissue so damage is concentrated (high RBE - weighting factor 20) **1** Long half-life means it continues to emit for the rest of the person's life / cannot be excreted quickly (contamination not just exposure) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	3

Do not write outside the box

Question	Marking guidance	Marks
01.5	1 mark for the point below. **1** A previously stable nucleus that becomes radioactive after absorbing a neutron (eg in reactor structures), later decaying to emit ionising radiation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 299-301; AQA spec 3.4.3.2.14.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** The climate naturally fluctuates (solar output, Earth's orbit) so extreme events can occur without human influence **1** An individual event, or even several, does not show a trend – the change may be in the frequency of events which can only be assessed years later **1** Changes occur at different time and spatial scales (local storm vs global trend) – interconnected systems make attribution complex **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Towns have a heat-island effect (heat emissions, dark surfaces absorbing light) **1** As the town grows around the weather station the warming recorded is partly local, not global – older equipment / sparse global coverage also reduce reliability **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**Marks shared** Up to 5 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Tree rings:** width of each annual ring shows the growth rate, indicating temperature / rainfall in that year – rings easily dated by counting – limitation: growth also affected by other factors (water, pests, shading), only covers the life of the tree / limited regions **up to 2 Ice cores:** annual snow layers compact into ice – air bubbles trapped in the ice give the CO2 (and methane) concentration when trapped – oxygen isotope ratios indicate temperature – up to 800 000 years (Dome C) – limitation: deep layers compressed / less distinct, dating needs radio-isotope analysis and volcanic ash layers – only polar regions **up to 2 Others accepted:** coral growth rings – pollen in lake sediments dated by radiocarbon **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 100-102; AQA spec 3.2.1.2.	
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Ploughing:** increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster – carbon in the soil DOM store falls and atmospheric CO2 rises **2 Draining:** waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly – drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO2 (though methane release stops) **2 Marine pollution:** kills phytoplankton so less dissolved CO2 is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton – less CO2 then dissolves from the atmosphere to replace it, so atmospheric CO2 rises **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air	6

Do not write outside the box

Question	Marking guidance	Marks
03.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Rice paddy fields **1** Landfill sites **1** Anaerobic sediments in reservoirs **1** Livestock intestines (cattle) **1** Manure / slurry stores **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
03.3	1 mark for the point below. **1** More CO_2 dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions – more H+ lowers the pH **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2.	
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) **1** Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (200 - 25) / 200 x 100 **1** = 87.5% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) **1** Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths **1** Higher doses caused sterility / death **1** After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) **1** Breeding success recovered so the population rose **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	4
04.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** DDT persists in soil / sediments / food chains for many years so exposure continued **1 Birds of prey are k-selected:** few young per year, slow to mature so populations grow slowly **1** Other pressures (habitat loss, persecution) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - rivers are accessible over a large area with few natural contaminants	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1 Potable / public supply:** no pathogens, low toxins, good appearance / taste / smell (public health) **1 Industrial boiler / cooling water:** soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange – spray irrigation: low turbidity to avoid blocked pipes and low heavy metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) **1 Disadvantage:** high cost / energy for pumping – ecological impacts on the donor river – transfer of pollutants or invasive species – political conflict **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable **1** Wild species have different ranges of tolerance so cress results cannot be applied directly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated • **tolerance** → the range of conditions in which a species can survive **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Preliminary study with widely spaced pH values to find where germination occurs **1** Final range from below the lowest pH giving germination to above the highest, with closer intervals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 4 points below. **1** Temperature (constant, natural germination temperature) **1** Light, water volume, number of seeds per dish, seed source **1** Growth medium (filter paper; soil would alter pH) – nutrients not needed for germination **1** Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface water affects fresh mass and drying damages the plants – dry mass requires killing the plants so the same plants cannot be re-measured **1** Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) **1** It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
07.2	**Any 4 of 8** 1 mark each for any 4 of the 8 points below, in any order, maximum 4. **1** Physical state of the effluent (solution / solid) **1** Ocean currents / river flow **1** Effect of pH and oxygen on solubility **1** Bioaccumulation and food-chain concentration **1** Food sources and consumption by local people (eg shellfish, seaweed) **1** Half-lives of the isotopes **1** Meteorological conditions for any gaseous release **1** Nature of the environment - geology, vegetation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain	4

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring focuses on the sites most at risk of contamination **1** No need to sample sites that were never likely to be polluted – sampling checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** It assesses the risk to the people most exposed because of their lifestyle before any health effects occur **1** If their risk is acceptably low everyone else has a lower risk – if unacceptable, emissions are reduced to prevent future exposure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Critical Group** → the people most at risk / most heavily exposed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
08.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Supply rate of energy crops can be controlled and fuels stored so supply matches demand (not intermittent) **1** Liquid biofuels (alcohol, biodiesel) can replace petrol / diesel in vehicles **1 Carbon neutral in principle:** CO2 released equals CO2 absorbed in photosynthesis **1** Energy density of alcohol / biodiesel nearly as high as fossil fuels – wastes (straw, sewage, landfill gas) are cheap **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous • **landfill gas** → the methane and carbon dioxide produced by anaerobic decomposition of waste **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Large areas of farmland compete with food production or cause clearance of wildlife habitats (eg palm oil) **1** Intensive farming inputs (fertiliser, pesticide, machinery fuel) may release nearly as much CO2 as using fossil fuels directly **1** Low energy density of straw, wood, Miscanthus makes transport uneconomic - must be used locally **1** Supply from wastes limited to the amount produced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
08.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Hydrogen from algae:** some algae deprived of sulfur produce hydrogen during photosynthesis **1** Anaerobic digestion extended to crop wastes and marine algae – small-scale digesters (eg one cow's manure supplying a family's cooking gas) **1** Biofuels from microorganisms (eg oil-producing microalgae) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Conventional geothermal steam turbines need rocks above 150 °C (recent volcanic activity), too deep to reach economically in the UK **1** New turbines use fluids such as butane / pentane that boil at low temperatures, so water as cool as 60 °C from deep boreholes / hot dry rock systems can drive them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 212-214; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** Ozone is very low overnight / early morning, rises from about 08:00, peaks at about 80 units at 14:00 and falls in the evening **1** Ozone is a secondary pollutant formed by photochemical reactions requiring sunlight (UV), so it peaks in the afternoon after several hours of sunlight **1** NO2 (from morning traffic) is broken down by UV to NO and monatomic oxygen – monatomic oxygen reacts with O2 to form ozone **1** Ozone lags behind the NO / NO2 peaks (which are produced by the morning rush-hour traffic) because the reactions take time – falls at night when there is no sunlight **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** NO is the primary pollutant produced in engines when nitrogen and oxygen react at high temperature / pressure **1** NO is oxidised in the atmosphere (by oxygen / ozone) to NO2, a secondary pollutant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** PANs (peroxyacetyl nitrates) **1** Eye irritation / breathing difficulty / asthma / emphysema / increased risk of heart attacks (toxic at much lower concentrations than NOx) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Heavy vehicle use giving high NOx and hydrocarbon emissions **1** Sunny climate providing the UV for photochemical reactions **1** Valley / basin topography and frequent temperature inversions trapping pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 248-249, 253-254; AQA spec 3.4.3.2.3-3.4.3.2.4.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Principles:** dB is logarithmic (10 dB = 10x; 3 dB = 2x) so small reductions in dB are large reductions in intensity – effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • **Aircraft design:** high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • **Aircraft operation:** take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • **Airport design / location:** away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • **Railway:** track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • **Road:** porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • **Industrial:** silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • **Domestic:** acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • **Evaluation:** which approaches are most effective / cost-effective – conflicts (airports near cities for convenience) – legislation and monitoring (Leq 57 dB contours) needed to enforce **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• Any appropriate named example, including ones that are not listed here. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Because the decibel scale is logarithmic, a 10 dB cut is a tenfold reduction in intensity and 3 dB halves it, so modest-sounding design changes matter. Aircraft engines have been made quieter by high-bypass-ratio turbofans, chevron nozzles, hush kits and acoustic liners, with fairings and composite airframes reducing airframe noise. Operation can be changed too: steeper take-off angles and flight paths that avoid housing, constant-descent approaches, restricted reverse thrust, night curfews, Chapter noise limits enforced by fines and quota counts. Airport location and layout - away from population centres, engine-test areas and taxiways kept from housing, acoustic walls and baffle mounds, land-use planning and free insulation for residents - deal with the noise that remains.	
	Railways use track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication on curves and composite brakes. Roads use porous or rubber-crumb asphalt, acoustic fences and embankments, tree planting, engine insulation and aerodynamics, traffic routing and double glazing. Industry fits silencers to compressors, presses rather than stamps, uses nylon rollers, drills instead of pile-driving, surrounds piling with bubble curtains to protect cetaceans, times blasting, hangs acoustic mats and operates remotely, with ear protection as a last line. Domestic appliances use acoustic absorbers and volume limiters.	
	Design changes at source are the most effective because they reduce noise for everyone; operational limits depend on enforcement; location works only for new developments, and airports are still sited near cities for convenience. Legislation and monitoring of noise contours (57 dB L_{eq}) are what turn these options into actual reductions.	
	Weaker answer - Level 2 (5 marks)	
	Noise pollution can be reduced in different ways. Aircraft can be designed with quieter engines and can fly at night less often. Airports can be built away from cities. Roads can have sound barriers next to them and quieter road surfaces. Trains can have their tracks polished and use better brakes. Factories can use quieter machines and workers can wear ear defenders. Noise can also be reduced by planting trees. These methods reduce the noise that reaches people.	
	What makes the difference	
	• Level 2 gives correct but generic examples ('quieter engines'); Level 3 names the design features (high bypass ratio, chevron nozzles, acoustic liners) the question's 'named examples' requires.	
	• Level 3 organises by design / operation / location as the question asks; Level 2 mixes them.	
	• The decibel-scale point and the closing evaluation of which approaches work best lift Level 3 into the top band.	
10.2	1 mark for the point below. **1** The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	
11.1	**Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • **Fossil fuels:** choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports – new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • **Nuclear:** high energy density reduces transport – waste properties (half-life) determine storage – thorium and fusion prospects • **Renewables:** matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting) – intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity – low-impact designs (helical turbines, VAWT, tidal lagoons) • **Energy conservation reduces demand:** transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use	25
	– life-cycle / embodied energy • **Evaluation:** no resource is impact-free – knowledge allows the best mix and mitigation – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

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Question	Marking guidance	Marks
11.2	**Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Problematic properties of solar:** intermittency, unreliability from cloud, low energy density, changing angle – locational constraints (deserts, latitude) • **Harnessing solar:** photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms – environmental impacts of manufacture and land use • **New solar technologies:** multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • **HEP:** gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links – combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	

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